

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	28-06-2026
Team ID	24A55A4203
Project Name	Visitor Management System
Maximum Marks	4 Marks

Technical Architecture:

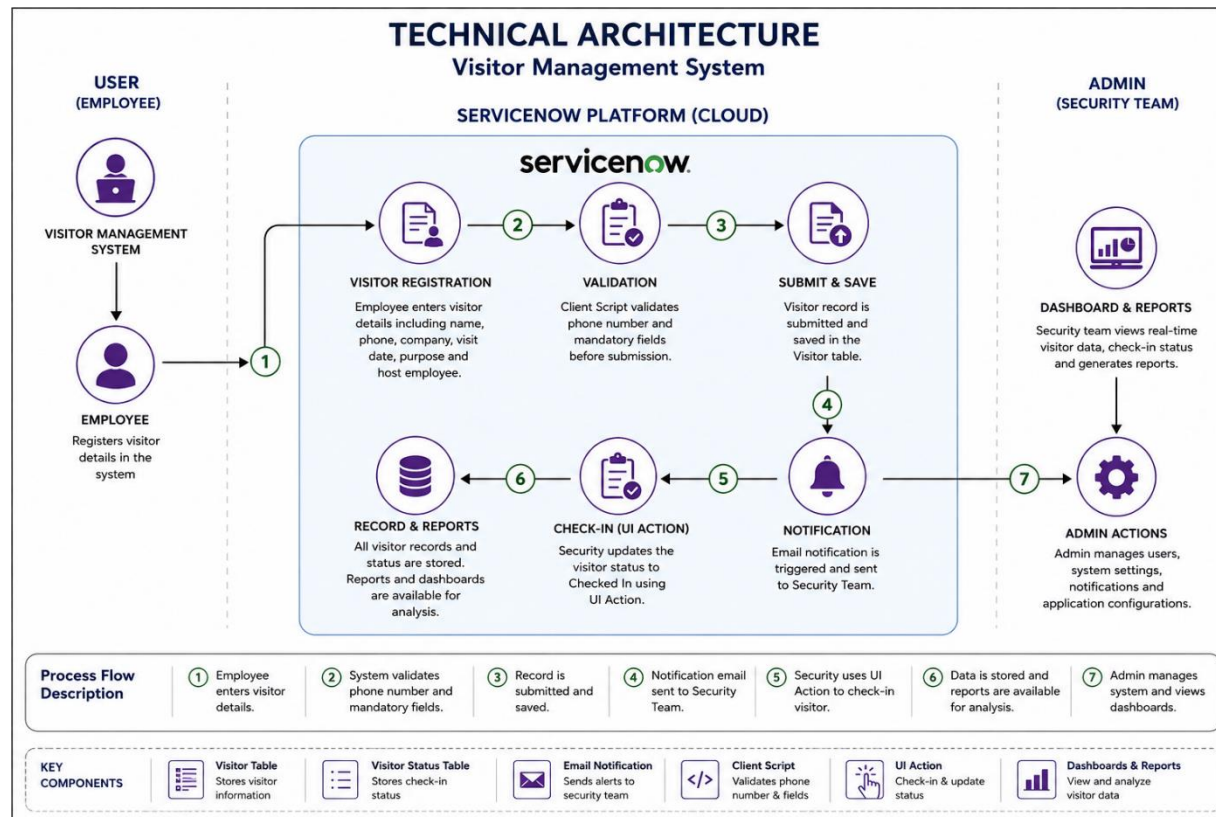


Table-1 : Components & Technologies:

S.No	Component	Description	Technology / Platform Layer
1	User Interface	Portal interfaces for employee pre-registration forms and front-desk kiosk interactions.	ServiceNow Service Portal, Portal UI
2	Application Logic-1	Client-side configuration rules verifying valid telephone formats or mandatory fields.	Catalog Client Scripts (JavaScript)
3	Application Logic-2	Server-side asynchronous retrieval of employee data and validation loops.	Script Includes, GlideAjax
4	Application Logic-3	Automated system workflow handling check-in triggers and logging operations.	Flow Designer, Business Rules
5	Database	Persistent internal table data structures tracking visitor logs, host relations, and times.	ServiceNow Tables (x_snc_visitor_log)
6	Cloud Database	SaaS cloud-based data tier managed natively by the enterprise platform.	ServiceNow Cloud Database
7	File Storage	Direct document processing storage hosting check-in passes or uploaded ID attachments.	ServiceNow Attachments (sys_attachment)
8	External API Integration	Dynamic verification generation engine or peripheral print utility integrations.	ServiceNow IntegrationHub / REST Messages
9	Notification Services	Transactional engine broadcasting automatic email alerts and SMS confirmations to hosts.	ServiceNow Notification Engine

10	Reporting & Intelligence	Analytical widgets providing security teams with real-time daily guest dashboards.	ServiceNow Reports & Dashboards
11	Infrastructure Host	Secure, resilient enterprise application hosting and multi-tenant runtime engine.	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

Application Attribute	Technology Mapping	Architecture Implementation Description
Open-Source Frameworks	JavaScript, GlideAjax	Client-side scripting architectures utilize native JavaScript frameworks to evaluate validation parameters without UI thread freezing.
Security Implementations	ServiceNow ACLs, User Roles, HTTPS, IAM Controls	Enforces strict row-level Access Control Lists (ACLs) to guarantee only designated security staff and admin contexts can read sensitive data logs.
Scalable Architecture	ServiceNow SaaS Multi-Tenant Cloud Platform	Platform scaling handles rapid spikes in lobby foot-traffic or massive increases in logged visitors without requiring infrastructure re-architecting.
High Availability	ServiceNow Cloud Managed Infrastructure	Ensures continuous operational uptime for security checkpoints through active redundant hosting nodes and continuous automated platform health monitoring.
High Performance	GlideAjax, Script Includes, Optimized DB Indexes	Asynchronous processing patterns prevent delays by querying database entries over optimized server-side logic scripts instead of locking the browser.